

SIMATS ENGINEERING

ASSESSMENT TOOL 2

**COURSE NAME: CRYPTOGRAPHY AND NETWORK
SECURITY COURSE CODE: CSA51**

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I Sweta R(192511408) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Sweta R

Annexure A

Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

Solution:

A. Explain why this issue occurs.

The company initially used **Electronic Codebook (ECB)** mode to encrypt financial data. In ECB mode, each plaintext block is encrypted independently using the same key. As a result, identical plaintext blocks always produce identical ciphertext blocks. Since financial transactions often contain repeated information, attackers can observe patterns in the ciphertext even though they cannot directly read the data. These visible patterns may reveal information about the structure of the original message, making ECB unsuitable for transmitting sensitive financial data.

B. Suggest a more secure block cipher mode and justify your choice.

A better choice is **Cipher Block Chaining (CBC)** mode. In CBC, each plaintext block is combined with the previous ciphertext block before encryption, and the first block uses an Initialization Vector (IV). This ensures that even if the same plaintext appears multiple times, the resulting ciphertext will be different. Consequently, CBC hides patterns in the encrypted data and provides much stronger confidentiality than ECB. Therefore, CBC is more suitable for securely transmitting repeated financial information.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish. .

A. Identify the most suitable algorithm for this scenario.

Among the available options, **Blowfish** is the most suitable algorithm for the banking application.

B. Justify your choice based on security strength and efficiency.

DES is no longer considered secure because its 56-bit key can be broken using modern computing power. Although 3-DES improves security by applying the DES algorithm three times, it is relatively slow and computationally expensive. Blowfish supports variable key lengths of up to 448 bits, providing stronger security while maintaining better performance than 3-DES. Since the banking system requires high security along with moderate performance for customer transactions, Blowfish offers the best balance between security.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

A. Explain how the key exchange works in this scenario.

Diffie-Hellman allows two users to establish a shared secret key over an unsecured public network without transmitting the secret key itself. Both users first agree on a common prime number and a generator, which can be made public. Each user then selects a private key and calculates a corresponding public key. After exchanging their public keys, each user combines the received public key with their own private key to generate the same shared secret. This shared secret can then be used as the encryption key for secure communication.

B. Identify a possible security threat and suggest a solution to mitigate it.

A major threat to Diffie-Hellman is the **Man-in-the-Middle (MITM) attack**. In this attack, an attacker intercepts the key exchange and establishes separate secret keys with each user while pretending to be the other party. As a result, the attacker can read or modify all transmitted messages. This threat can be prevented by authenticating the communicating users using digital certificates, digital signatures, or Public Key Infrastructure (PKI). Authentication ensures that the exchanged public keys actually belong to the intended users.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

A. Describe how RSA can be used for encryption and decryption in this scenario.

RSA uses a pair of mathematically related keys: a public key and a private key. When sending a confidential email, the sender encrypts the message using the recipient's public key. Since only the recipient possesses the corresponding private key, only they can decrypt and read the message. For authentication, the sender can also create a digital signature using their private key. The recipient verifies the signature using the sender's public key, ensuring that the email is authentic and has not been modified during transmission.

B. Explain how key distribution and management play a role in maintaining security.

The security of RSA depends on proper key management. Public keys must be distributed through trusted sources to prevent attackers from replacing them with fake keys. Private keys must be securely stored and never shared with others. Organizations should also implement policies for key generation, secure storage, backup, periodic renewal, and revocation of compromised keys. Effective key management ensures confidentiality, authenticity, and

integrity of email communications

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

A. Analyze the constraints of IoT devices.

IoT devices such as smart sensors and home automation systems typically have limited processing power, memory, storage capacity, and battery life. These devices often operate continuously and must perform encryption without consuming excessive energy. Therefore, cryptographic algorithms designed for IoT must provide strong security while requiring minimal computational resources.

B. Recommend a suitable cryptographic approach and justify your answer.

Elliptic Curve Cryptography (ECC) is the most suitable choice for IoT devices. ECC provides a security level comparable to RSA while using much smaller key sizes. Smaller keys require less memory, lower processing power, and reduced energy consumption, making ECC highly efficient for low-power devices. It also enables faster key generation and secure communication without placing significant demands on hardware resources. For these reasons, ECC is widely adopted in modern IoT systems and smart home applications.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well reasoned, logical decisions with strong	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided

		rationale			
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand